

CBSE Class 12 Maths — Matrices and Determinants

50 Most-Probable PYQs & Board-Pattern Practice Questions

1-Mark (MCQ/Assertion-Reason/VSA)

Q1. Construct a 2×2 matrix $A = [a_{ij}]$ whose elements are given by $a_{ij} = \frac{(i+2j)^2}{2}$. [PYQ 2019] [Confidence: High]

Q2. If A is a square matrix of order 3 and $|A| = 5$, find the value of $|\text{adj}A|$. [PYQ 2023] [Confidence: Very High]

Q3. Find the values of x and y if $\begin{pmatrix} 2x+y & 4 \\ 5 & x-2y \end{pmatrix} = \begin{pmatrix} 7 & 4 \\ 5 & 1 \end{pmatrix}$. [PYQ 2017] [Confidence: High]

Q4. If A is a skew-symmetric matrix of order 3, what is the value of its determinant $|A|$? [PYQ 2020] [Confidence: Very High]

Q5. For what value of x is the matrix $A = \begin{pmatrix} 5-x & x+1 \\ 2 & 4 \end{pmatrix}$ a singular matrix? [PYQ 2018] [Confidence: High]

Q6. If $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix}$, find the value of the determinant $|AB|$. [Practice] [Confidence: Medium]

Q7. Given A is a square matrix of order 3 and $|3A| = k|A|$, find the value of k . [PYQ 2022] [Confidence: Very High]

Q8. Write the maximum possible value of a 2×2 determinant whose elements are chosen from the set $\{0, 1\}$. [PYQ 2016] [Confidence: Medium]

Q9. If A is a square matrix such that $A^2 = I$, find the simplified value of $(A - I)^3 + (A + I)^3 - 7A$. [PYQ 2019] [Confidence: High]

Q10. If A and B are symmetric matrices of the same order, then what is the nature of the matrix $AB - BA$? [PYQ 2021] [Confidence: Very High]

Q11. If $\begin{vmatrix} 2x & 5 \\ 8 & x \end{vmatrix} = \begin{vmatrix} 6 & -2 \\ 7 & 3 \end{vmatrix}$, find the value of x . [PYQ 2018] [Confidence: High]

Q12. Find the cofactor of the element a_{12} in the determinant $\begin{vmatrix} 2 & -3 & 5 \\ 6 & 0 & 4 \\ 1 & 5 & -7 \end{vmatrix}$. [Practice] [Confidence: Medium]

Q13. Write the number of all possible matrices of order 3×3 with each entry being 0 or 1. [PYQ 2017] [Confidence: Very High]

Q14. If A is a non-singular square matrix of order 3 such that $A^2 = 3A$, find the value of $|A|$. [Practice] [Confidence: High]

Q15. Find the adjoint of the matrix $A = \begin{pmatrix} 2 & -1 \\ 4 & 3 \end{pmatrix}$. [PYQ 2020] [Confidence: High]

Q16. A system of linear equations $AX = B$ is said to be inconsistent if $|A| = 0$ and $(\text{adj}A)B$ is equal to what? [Practice] [Confidence: Medium]

Q17. What is the trace of the matrix $A = \begin{pmatrix} 1 & -2 & 3 \\ 4 & 5 & -6 \\ 7 & 8 & 9 \end{pmatrix}$? [Practice] [Confidence: Medium]

Q18. Assertion (A): The inverse of a matrix exists if and only if it is non-singular. **Reason (R):** For a non-singular matrix A , $|A| \neq 0$. Choose the correct option. [PYQ 2024] [Confidence: Very High]

Q19. Assertion (A): For any two invertible matrices A and B , $(AB)^{-1} = A^{-1}B^{-1}$. **Reason (R):** Matrix multiplication is generally not commutative. Choose the correct option. [PYQ 2024] [Confidence: High]

Q20. Assertion (A): The determinant of a diagonal matrix is the product of its principal diagonal elements. **Reason (R):** All non-diagonal elements of a diagonal matrix are zero. Choose the correct option. [Practice] [Confidence: High]

2-Mark

Q21. If $A = \begin{pmatrix} 2 & 3 \\ 4 & -1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & -2 \\ 3 & 5 \end{pmatrix}$, find a matrix X such that $2A + 3X = 5B$. [PYQ 2018] [Confidence: High]

Q22. If $A = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}$, then for what value of α lying in $[0, \frac{\pi}{2}]$ is $A + A^T = I$? [PYQ 2020] [Confidence: Very High]

Q23. Using determinants, find the value of k if the area of the triangle with vertices $(2, -6)$, $(5, 4)$, and $(k, 4)$ is 35 sq. units. [PYQ 2022] [Confidence: High]

Q24. Show that the points $A(a, b + c)$, $B(b, c + a)$, and $C(c, a + b)$ are collinear using determinants. [PYQ 2017] [Confidence: High]

Q25. If $A = \begin{pmatrix} 3 & 1 \\ -1 & 2 \end{pmatrix}$, show that $A^2 - 5A + 7I = 0$. [PYQ 2019] [Confidence: Very High]

Q26. Find the matrix X such that $X \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 4 & 5 \\ 0 & 1 \end{pmatrix}$. [PYQ 2018] [Confidence: High]

Q27. Find the inverse of the matrix $A = \begin{pmatrix} 2 & -3 \\ -1 & 2 \end{pmatrix}$ using the formula $A^{-1} = \frac{1}{|A|} \text{adj}A$. [Practice] [Confidence: Medium]

Q28. If $A = \begin{pmatrix} 1 & 5 \\ 6 & 7 \end{pmatrix}$, verify that $A + A^T$ is a symmetric matrix. [PYQ 2021] [Confidence: High]

Q29. Let A be a non-singular matrix of order 3×3 . If $|A| = 4$, find the exact values of $|A^{-1}|$ and $|A \cdot \text{adj}A|$. [Practice] [Confidence: Very High]

Q30. Examine the consistency of the system of equations: $x + 2y = 2$ and $2x + 3y = 3$. [PYQ 2020] [Confidence: Medium]

Q31. Evaluate the determinant: $\begin{vmatrix} \sin^2 \theta & \cos^2 \theta \\ -\cos^2 \theta & \sin^2 \theta \end{vmatrix}$. [Practice] [Confidence: High]

Q32. If A is a skew-symmetric matrix of order 3, prove algebraically that its determinant is zero. [PYQ 2016] [Confidence: Medium]

3-Mark

Q33. Express the matrix $A = \begin{pmatrix} 3 & 5 \\ 1 & -1 \end{pmatrix}$ as the sum of a symmetric and a skew-symmetric matrix. [PYQ 2020] [Confidence: Very High]

Q34. If $f(x) = x^2 - 4x + 1$, find $f(A)$ when $A = \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}$ and show that $f(A)$ is a null matrix. [PYQ 2018] [Confidence: High]

Q35. Find the equation of the line joining points $A(1, 3)$ and $B(0, 0)$ using determinants, and find k if $D(k, 0)$ is a point such that the area of $\triangle ABD$ is 3 sq units. [PYQ 2023] [Confidence: High]

Q36. If $A = \begin{pmatrix} 1 & -1 & 2 \\ 3 & 0 & -2 \\ 1 & 0 & 3 \end{pmatrix}$, compute $\text{adj}A$ and verify that $A(\text{adj}A) = |A|I$. [PYQ 2019] [Confidence: Very High]

Q37. Prove that $A^n = \begin{pmatrix} 1+2n & -4n \\ n & 1-2n \end{pmatrix}$ where $A = \begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix}$ for all $n \in \mathbb{N}$, using the principle of mathematical induction. [PYQ 2017] [Confidence: Medium]

Q38. If $A = \begin{pmatrix} 2 & -1 \\ 3 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 4 \\ -1 & 7 \end{pmatrix}$, find $(AB)^{-1}$. [Practice] [Confidence: High]

Q39. Evaluate the determinant: $\begin{vmatrix} 1 & a & b+c \\ 1 & b & c+a \\ 1 & c & a+b \end{vmatrix}$. [PYQ 2022] [Confidence: Very High]

Q40. For what values of a and b does the system of equations $2x + 3y = 7$ and $2ax + (a + b)y = 28$ have infinitely many solutions? [Practice] [Confidence: Medium]

Q41. Find matrix A such that $\begin{pmatrix} 2 & -1 \\ 1 & 0 \\ -3 & 4 \end{pmatrix} A = \begin{pmatrix} -1 & -8 \\ 1 & -2 \\ 9 & 22 \end{pmatrix}$. [PYQ 2019] [Confidence: High]

Q42. Show that a matrix which is both symmetric and skew-symmetric must be a zero matrix. [Practice] [Confidence: Medium]

5-Mark (Long Answer/Case-Study)

Q43. Using the matrix method, solve the following system of linear equations: $2x + 3y + 3z = 5$ $x - 2y + z = -4$ $3x - y - 2z = 3$ [PYQ 2020] [Confidence: Very High]

Q44. If $A = \begin{pmatrix} 2 & -3 & 5 \\ 3 & 2 & -4 \\ 1 & 1 & -2 \end{pmatrix}$, find A^{-1} . Using A^{-1} , solve the system of equations: $2x - 3y + 5z = 11$

$3x + 2y - 4z = -5$ $x + y - 2z = -3$ [PYQ 2023] [Confidence: Very High]

Q45. Use the product $\begin{pmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 3 & -2 & 4 \end{pmatrix} \begin{pmatrix} -2 & 0 & 1 \\ 9 & 2 & -3 \\ 6 & 1 & -2 \end{pmatrix}$ to solve the system of equations: $x - y + 2z = 1$

$2y - 3z = 1$ $3x - 2y + 4z = 2$ [PYQ 2019] [Confidence: Very High]

Q46. Solve the following system of equations using the matrix method: $\frac{2}{x} + \frac{3}{y} + \frac{10}{z} = 4$ $\frac{4}{x} - \frac{6}{y} + \frac{5}{z} = 1$ $\frac{6}{x} + \frac{9}{y} - \frac{20}{z} = 2$ [PYQ 2018] [Confidence: High]

Q47. Case Study: Three shopkeepers Ram Lal, Shyam Lal, and Ghansham are using polythene bags, handmade bags (prepared by prisoners), and newspaper envelopes as carrying bags. It is found that the shopkeepers use (20, 30, 40), (30, 40, 20), and (40, 20, 30) polythene bags, handmade bags, and newspaper envelopes respectively. Ram Lal, Shyam Lal, and Ghansham spent ₹250, ₹270, and ₹200 on these carry bags respectively. (i) Express the above situation as a matrix equation. (2 Marks) (ii) Find the cost of one polythene bag, one handmade bag, and one newspaper envelope using the matrix method. (3 Marks) [PYQ 2022] [Confidence: High]

Q48. Case Study: The cost of 4 kg onion, 3 kg wheat, and 2 kg rice is ₹60. The cost of 2 kg onion, 4 kg wheat, and 6 kg rice is ₹90. The cost of 6 kg onion, 2 kg wheat, and 3 kg rice is ₹70. (i) Represent the given data in the form of a system of linear equations and formulate it into a matrix equation $AX = B$. (1 Mark) (ii) Determine the determinant of the coefficient matrix A . (1 Mark) (iii) Find the cost of each item per kg by solving the matrix equation. (3 Marks) [PYQ 2021] [Confidence: Very High]

Q49. If $A = \begin{pmatrix} 1 & 2 & 0 \\ -2 & -1 & -2 \\ 0 & -1 & 1 \end{pmatrix}$, find A^{-1} . Using A^{-1} , solve the system of linear equations: $x - 2y = 10$

$2x - y - z = 8$ $-2y + z = 7$ [PYQ 2017] [Confidence: High]

Q50. Case Study: Three schools A, B, and C decided to organize a fair for collecting funds for awarding their students for the core values of Honesty, Punctuality, and Discipline. School A awards ₹ x , ₹ y , and ₹ z to 3, 2, and 1 students respectively, distributing a total of ₹1000. School B awards 4, 1, and 3 students respectively for the same values, distributing ₹1500. School C awards 2, 3, and 1 students respectively, distributing ₹1200. (i) Formulate this scenario into a matrix equation $AX = B$. (2 Marks) (ii) Using the matrix inverse method, find the exact award amount for Honesty, Punctuality, and Discipline. (3 Marks) [Practice] [Confidence: High]

Answer Key

Q1. $a_{11} = \frac{9}{2}, a_{12} = \frac{25}{2}, a_{21} = 8, a_{22} = 18 \Rightarrow A = \begin{pmatrix} 9/2 & 25/2 \\ 8 & 18 \end{pmatrix}$. **Q2.** $|\text{adj}A| = |A|^{n-1} = 5^{3-1} = 5^2 = 25$.

Q3. Solving $2x + y = 7$ and $x - 2y = 1$, we get $x = 3, y = 1$. **Q4.** $|A| = 0$ (Determinant of a skew-symmetric matrix of odd order is always 0). **Q5.**

$|A| = 0 \Rightarrow 4(5 - x) - 2(x + 1) = 0 \Rightarrow 20 - 4x - 2x - 2 = 0 \Rightarrow 6x = 18 \Rightarrow x = 3$. **Q6.**

$|A| = (4 - 6) = -2, |B| = (6 - 0) = 6. |AB| = |A||B| = (-2)(6) = -12$. **Q7.** $|3A| = 3^3|A| = 27|A|$, so

$k = 27$. **Q8.** 1 (e.g., matrix with rows (1, 0) and (0, 1)). **Q9.** Expand: $(A - I)^3 + (A + I)^3 - 7A$. Note

$A^2 = I, A^3 = A$. Using binomial expansion, it simplifies to $2A^3 + 6AI^2 - 7A = 2A + 6A - 7A = A$. **Q10.** Skew-symmetric matrix. $(AB - BA)^T = B^T A^T - A^T B^T = BA - AB = -(AB - BA)$. **Q11.**

$2x^2 - 40 = 18 - (-14) \Rightarrow 2x^2 - 40 = 32 \Rightarrow 2x^2 = 72 \Rightarrow x = \pm 6$. **Q12.** Cofactor

$C_{12} = (-1)^{1+2} M_{12} = -[6(-7) - 4(1)] = -[-42 - 4] = 46$. **Q13.** Total entries = 9. Options per entry = 2. Total matrices = $2^9 = 512$. **Q14.** $|A^2| = |3A| \Rightarrow |A|^2 = 3^3|A| = 27|A|$. Since $|A| \neq 0, |A| = 27$. **Q15.** Swap diagonal

elements and change sign of non-diagonal elements: $\text{adj}A = \begin{pmatrix} 3 & 1 \\ -4 & 2 \end{pmatrix}$. **Q16.** A non-zero matrix, i.e.,

$(\text{adj}A)B \neq O$. **Q17.** Trace = Sum of principal diagonal elements = $1 + 5 + 9 = 15$. **Q18.** Option A (Both A and R are true, and R is the correct explanation of A). **Q19.** Option D (A is false because $(AB)^{-1} = B^{-1}A^{-1}$, but R is true).

Q20. Option A (Both A and R are true, and R is the correct explanation of A).

Q21. $X = \frac{1}{3}(5B - 2A) = \frac{1}{3} \left[\begin{pmatrix} 5 & -10 \\ 15 & 25 \end{pmatrix} - \begin{pmatrix} 4 & 6 \\ 8 & -2 \end{pmatrix} \right] = \begin{pmatrix} 1/3 & -16/3 \\ 7/3 & 9 \end{pmatrix}$. **Q22.**

$A + A^T = \begin{pmatrix} 2\cos\alpha & 0 \\ 0 & 2\cos\alpha \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \Rightarrow 2\cos\alpha = 1 \Rightarrow \alpha = \frac{\pi}{3}$. **Q23.**

$\frac{1}{2} \begin{vmatrix} 2 & -6 & 1 \\ 5 & 4 & 1 \\ k & 4 & 1 \end{vmatrix} = \pm 35 \Rightarrow \frac{1}{2}[2(0) + 6(5 - k) + 1(20 - 4k)] = \pm 35 \Rightarrow 50 - 10k = \pm 70 \Rightarrow k = -2, 12$

Q24. Evaluate area of triangle determinant: applying $C_2 \rightarrow C_2 + C_1$ gives $(a + b + c)$ in C_2 , making C_2

proportional to C_3 . Area = 0. **Q25.** $A^2 = \begin{pmatrix} 8 & 5 \\ -5 & 3 \end{pmatrix}, 5A = \begin{pmatrix} 15 & 5 \\ -5 & 10 \end{pmatrix}$.

$A^2 - 5A + 7I = \begin{pmatrix} 8 - 15 + 7 & 5 - 5 + 0 \\ -5 + 5 + 0 & 3 - 10 + 7 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$. **Q26.** Let $X = \begin{pmatrix} x & y \\ z & w \end{pmatrix}$. Solving

$X \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 4 & 5 \\ 0 & 1 \end{pmatrix} \Rightarrow X = \begin{pmatrix} 4 & 5 \\ 0 & 1 \end{pmatrix} \frac{1}{-2} \begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} -1/2 & 1 \\ -3/2 & 1 \end{pmatrix}$. **Q27.** $|A| = 4 - 3 = 1$.

$\text{adj}A = \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}, A^{-1} = \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}$. **Q28.** $A + A^T = \begin{pmatrix} 1 & 5 \\ 6 & 7 \end{pmatrix} + \begin{pmatrix} 1 & 6 \\ 5 & 7 \end{pmatrix} = \begin{pmatrix} 2 & 11 \\ 11 & 14 \end{pmatrix}$. Transpose is the same, so it is symmetric. **Q29.** $|A^{-1}| = \frac{1}{|A|} = \frac{1}{4}$. $|A \cdot \text{adj}A| = ||A|I| = |A|^3 = 4^3 = 64$. **Q30.**

$|A| = \begin{vmatrix} 1 & 2 \\ 2 & 3 \end{vmatrix} = 3 - 4 = -1 \neq 0$. Therefore, the system is consistent with a unique solution. **Q31.**

$\sin^2 \theta (\sin^2 \theta) - (-\cos^2 \theta)(\cos^2 \theta) = \sin^4 \theta + \cos^4 \theta = 1 - 2\sin^2 \theta \cos^2 \theta$. **Q32.** Let $A^T = -A$.

$|A^T| = |-A| \Rightarrow |A| = (-1)^3|A| \Rightarrow |A| = -|A| \Rightarrow 2|A| = 0 \Rightarrow |A| = 0$.

Q33. $A = P + Q$ where $P = \frac{1}{2}(A + A^T) = \begin{pmatrix} 3 & 3 \\ 3 & -1 \end{pmatrix}$ and $Q = \frac{1}{2}(A - A^T) = \begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix}$. **Q34.**

$A^2 = \begin{pmatrix} 7 & 12 \\ 4 & 7 \end{pmatrix}, f(A) = A^2 - 4A + I = \begin{pmatrix} 7 & 12 \\ 4 & 7 \end{pmatrix} - \begin{pmatrix} 8 & 12 \\ 4 & 8 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$. **Q35.** Equation of line:

$\frac{1}{2} \begin{vmatrix} x & y & 1 \\ 1 & 3 & 1 \\ 0 & 0 & 1 \end{vmatrix} = 0 \Rightarrow 3x - y = 0$. Area ΔABD : $\frac{1}{2} \begin{vmatrix} 1 & 3 & 1 \\ 0 & 0 & 1 \\ k & 0 & 1 \end{vmatrix} = \pm 3 \Rightarrow \frac{1}{2}(-3k) = \pm 3 \Rightarrow k = \pm 2$. **Q36.**

$|A| = 1(0) - (-1)(11) + 2(0) = 11$. Find cofactors, transpose to get $\text{adj}A$. Then multiply $A \cdot \text{adj}A$ to verify it

equals $11 \cdot I$. **Q37.** Show true for $n = 1$: $A^1 = \begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix}$. Assume true for $n = k$. Prove for $n = k + 1$:

$A^{k+1} = A^k \cdot A$. Multiply the assumed formula by A to reach the $(k + 1)$ result. **Q38.** Find $A^{-1} = \frac{1}{7} \begin{pmatrix} 2 & 1 \\ -3 & 2 \end{pmatrix}$.

$B^{-1} = \frac{1}{4} \begin{pmatrix} 7 & -4 \\ 1 & 0 \end{pmatrix}$. Use $(AB)^{-1} = B^{-1}A^{-1} = \frac{1}{28} \begin{pmatrix} 26 & -1 \\ 2 & 1 \end{pmatrix}$. **Q39.** Use $C_3 \rightarrow C_3 + C_2$. C_3 becomes

$(a+b+c)$ which can be taken out. C_3 becomes identically 1, matching C_1 . Value is 0. **Q40.** For infinite solutions, $\frac{2}{2a} = \frac{3}{a+b} = \frac{7}{28}$. From $\frac{2}{2a} = \frac{1}{4} \Rightarrow a = 4$. From $\frac{3}{4+b} = \frac{1}{4} \Rightarrow b = 8$. **Q41.** Let A be a 2×2 matrix. By multiplying rows with $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, we get $a = 1, b = -2, c = 3, d = 4 \Rightarrow A = \begin{pmatrix} 1 & -2 \\ 3 & 4 \end{pmatrix}$. **Q42.** $A^T = A$ and $A^T = -A$. Therefore, $A = -A \Rightarrow 2A = 0 \Rightarrow A = 0$.

Q43. $A = \begin{pmatrix} 2 & 3 & 3 \\ 1 & -2 & 1 \\ 3 & -1 & -2 \end{pmatrix}$, $|A| = 40$. $A^{-1} = \frac{1}{40} \begin{pmatrix} 5 & 3 & 9 \\ 5 & -13 & 1 \\ 5 & 11 & -7 \end{pmatrix}$. $X = A^{-1}B \Rightarrow x = 1, y = 2, z = -1$.

Q44. $|A| = -1$. $A^{-1} = \begin{pmatrix} 0 & 1 & -2 \\ -2 & 9 & -23 \\ -1 & 5 & -13 \end{pmatrix}$. However, the coefficients of the equations form A^T . The solution is

$X = (A^T)^{-1}B = (A^{-1})^T B \Rightarrow x = 1, y = 2, z = 3$. **Q45.** Product evaluates to $17I$. Thus

$A \cdot B = 17I \Rightarrow A^{-1} = \frac{1}{17}B$. The given system coefficients match A . So

$X = A^{-1}C = \frac{1}{17}B \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} \Rightarrow x = 0, y = 5/17, z = 3/17$. **Q46.** Let $1/x = u, 1/y = v, 1/z = w$. Solve

$2u + 3v + 10w = 4$, etc. $|A| = 1200$. Solve to get $u = 1/2, v = 1/3, w = 1/5 \Rightarrow x = 2, y = 3, z = 5$. **Q47.** (i)

$\begin{pmatrix} 20 & 30 & 40 \\ 30 & 40 & 20 \\ 40 & 20 & 30 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 250 \\ 270 \\ 200 \end{pmatrix}$. (ii) Solve $AX = B$ using A^{-1} . $x = 1, y = 5, z = 2$. **Q48.** (i)

$\begin{pmatrix} 4 & 3 & 2 \\ 2 & 4 & 6 \\ 6 & 2 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 60 \\ 90 \\ 70 \end{pmatrix}$. (ii) $|A| = 50$. (iii) Solving $X = A^{-1}B \Rightarrow x = 5, y = 8, z = 8$. **Q49.** $|A| = -1$.

$A^{-1} = \begin{pmatrix} -3 & -2 & -4 \\ 2 & 1 & 2 \\ 2 & 1 & 3 \end{pmatrix}$. System matrix is transpose of A .

$X = (A^T)^{-1}B = (A^{-1})^T B \Rightarrow x = 0, y = -5, z = -3$. **Q50.** (i) $\begin{pmatrix} 3 & 2 & 1 \\ 4 & 1 & 3 \\ 2 & 3 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1000 \\ 1500 \\ 1200 \end{pmatrix}$. (ii) Solve to

find $x = 200, y = 100, z = 200$.